Provider-Dependent Reliability of Self-Reported Confidence under a Shared Black-Box MT Protocol

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Abstract

This paper studies a trustworthy-AI reliability question for black-box language-model services: can one shared confidence threshold be reused across providers under a common protocol? We evaluate English→German translation with eight models from OpenAI, Anthropic, and Google on 500 WMT17 sentences (4,000 translation–confidence pairs) at temperature 0. Quality is scored with sentence-level chrF, and low quality is operationally defined as the within-model bottom 20% tail. Relative to that target, expected calibration error (ECE) and mismatch@0.9 differ widely despite similar mean chrF (54.0–60.1): ECE ranges from 0.076 to 0.180, and mismatch@0.9 ranges from 0.2% to 18.2%. The released snapshot also shows uneven parse-warning burden (443/4000 rows; 11.1%), an implementation-level confound for cross-provider reliability comparisons. The central claim is therefore practical and scoped: under this shared protocol and operational target, self-reported confidence is provider-dependent and non-portable as a common probability scale, so a fixed threshold does not induce comparable routing/abstention operating points. Supplementary analyses indicate that provider-specific isotonic remapping can reduce ECE but does not restore portability. Limitations are explicit: operational (not semantic) target, no bundled COMET/xCOMET checkpoints, no completed semantic-audit labels, and no bundled non-baseline prompt-variant outputs.

1 Introduction

Large language models (LLMs) are increasingly used as machine translation systems. In many deployment settings, average translation quality is not enough: a usable system also needs a signal that helps decide when an output should be trusted, routed for review, or deferred. For closed-source APIs, prompting the model to output a scalar confidence is attractive because it is simple, model-agnostic, and inexpensive. Recent work on LLM confidence estimation, however, shows that prompt-elicited or verbalized confidence is often miscalibrated, prompt-sensitive, and difficult to compare across settings.

This paper studies that problem in a controlled English-to-German MT setting and focuses on a narrower question than general confidence estimation: whether prompt-elicited, self-reported confidence is comparable enough across providers to support fixed decision policies such as abstention or human routing. This is a practically relevant question because deployment policies often treat confidence as a shared scalar interface: if different providers emit values in the same nominal range $[0, 1]$, it is natural to ask whether a common threshold can be reused across systems under a shared prompting protocol. In the released artifact, canonical prompts are rendered by shared prompt-variant functions and then passed through provider adapters, so provider comparisons are based on the same canonical wording and JSON schema. For each source sentence, the pipeline collects (i) a translation and (ii) a post-hoc confidence score conditioned on the source and the model output. Confidence is then evaluated against a reproducible reference-based quality signal and, secondarily, against a lightweight source-side surface-complexity proxy.

The key construct in this paper is operational rather than semantic. The analysis does not use human judgments of correctness. Instead, it defines a low-quality event through sentence-level chrF quantiles and asks whether self-reported confidence tracks that event under a shared protocol. The resulting comparisons are therefore about the portability of confidence relative to a reproducible chrF-based target, not about direct calibration to semantic correctness. This distinction matters throughout: the paper is strongest when read as a comparison of fixed-threshold operating points across providers rather than as a complete account of confidence utility.

The analysis addresses two questions:

1. **Comparative alignment under an operational low-quality definition:** how do self-reported confidence scores align with a reproducible chrF-based low-quality event when that event is defined by quantiles?
2. **Provider portability under a fixed threshold:** how often do high-confidence low-quality outputs occur under a shared threshold, and does this rate vary across source-side surface-complexity strata?

The main result is practical rather than theoretical. Models with similar mean chrF differ substantially in ECE and in the rate at which they assign high confidence to low-quality outputs at strict thresholds. A fixed policy such as "route to a human if confidence < 0.9" is therefore not portable across providers without calibration or threshold tuning. At the same time, low mismatch at a strict threshold does not by itself prove superior selective utility, because a conservative model can also look safer simply by rarely emitting values above that threshold. The surface-complexity analysis is more modest: it is included as a secondary stratification analysis rather than as a validated model of translation difficulty.

Contributions.

- A reproducible black-box evaluation pipeline for provider-level confidence reliability comparisons under a shared MT protocol.
- Evidence that fixed confidence thresholds induce materially different operating points across providers, even when mean translation quality is similar.
- A conservative deployment-oriented framing for trustworthy decision support: confidence-guided routing or abstention should be calibrated per provider, with explicit caveats about operational labels, parse burden, and incomplete semantic validation.

2 Related work

MT evaluation, quality estimation, and reliability signals. Automatic MT evaluation has long relied on overlap-based metrics such as BLEU and its standardized variants, as well as character-level metrics such as chrF. Learned metrics, including COMET and BLEURT, generally correlate better with human judgments. More recent work has continued to strengthen this line: xCOMET adds fine-grained error detection on top of sentence-level scoring, and the WMT24 Metrics Shared Task specifically examined how metrics behave on LLM-generated translations. Quality estimation (QE) predicts translation quality without references and remains an active WMT shared-task line. MT confidence estimation itself is also a longstanding topic rather than a new LLM-era concern: early work studied sentence-level and word-level confidence estimation for MT outputs well before contemporary API-based systems [1]. This paper does

not propose a new QE or confidence-estimation model; it studies a narrower reliability signal emitted by the generating model itself.

LLM-based MT evaluation and uncertainty-aware MT reliability. Recent work has also studied whether LLMs can serve as MT evaluators and what information they need to do so reliably. This literature is relevant because it frames translation reliability as a prediction problem, but it typically studies external evaluation signals rather than self-reported confidence from the model that produced the translation. Closer in spirit are confidence-aware MT reliability approaches that attach uncertainty estimates to translation quality prediction, including recent confidence-aware QE models such as Early-Exit and Instant Confidence COMET. Relative to that line, the present paper asks a narrower black-box question: whether the self-reported confidence already exposed through a shared API-style interaction is stable enough across providers to support common threshold policies.

Uncertainty, calibration, and selective prediction. Modern neural networks are often miscalibrated. Calibrated probabilities are useful for selective prediction or abstention policies, where a system chooses when to defer. Classical approaches estimate uncertainty through approximate Bayesian methods such as MC dropout or deep ensembles, followed by post-hoc calibration. For LLMs, the literature has expanded rapidly: recent surveys synthesize the challenges of confidence estimation and calibration for generated text, especially in black-box settings where access to logits is unavailable or impractical. Black-box calibration methods that use only textual generations are particularly relevant here because they match the interface available in many provider APIs [2].

Verbalized confidence and self-evaluation in LLMs. Prior work suggests that language models can sometimes assess the validity of their own outputs when asked in the right format. More recent work studies verbalized confidence directly: Yang et al. show that such scores can be useful but are highly prompt-dependent, while Wang et al. analyze how post-hoc calibration can improve verbalized probability estimates for black-box models. Yona et al. further show that models often fail to express intrinsic uncertainty faithfully in words, which is directly relevant when a scalar confidence is elicited through natural-language instructions rather than read from model internals [3]. In multilingual settings, MlingConf shows that confidence estimation itself can vary across languages and prompting conditions.

Relative to QE, uncertainty-aware MT evaluation, and LLM-as-evaluator work, the present paper studies a narrower but practically distinct signal: the self-reported confidence emitted by the same black-box model that produced the translation. That signal is weaker than an external evaluator in principle, but it is attractive in deployment because it is immediate, cheap, and provider-agnostic at the API level. The question addressed here is therefore not whether self-reported confidence is the best possible reliability estimator, but whether it is stable enough across providers to support shared threshold policies under a single prompt protocol. The results suggest that, under the protocol and target used here, it is not.

3 Experimental setup

3.1 Dataset

The experiments use the WMT17 English-to-German test set, accessed through `sacrebleu`. From the full test set, 500 sentence pairs are sampled with a fixed random seed (123) and stored as JSONL records containing `id`, `src`, and `ref`. A 500-sentence sample was chosen to keep the

multi-provider evaluation financially and operationally tractable while still producing 4,000 total model outputs for comparative analysis.

3.2 Models and providers

Eight LLMs across three providers are evaluated: OpenAI (gpt-5.2, gpt-5-mini, gpt-5-nano), Anthropic (claude-opus-4-6, claude-sonnet-4-5-20250929, claude-haiku-4-5-20251001), and Google (gemini-2.5-pro, gemini-2.5-flash). Each model is queried on the same 500 inputs. In the canonical artifact snapshot used here, all models contribute 500 usable confidence values for 500 total inputs, as reported in Table 1. The accompanying materials record the exact provider model identifiers used in the canonical runs so that the reported comparisons remain tied to a specific released snapshot rather than to shifting provider aliases in the abstract.

3.3 Prompting protocol

Each input sentence triggers two calls:

1. **Translation:** English-to-German translation, returned as a strict JSON object with key `translation`.
2. **Confidence:** a scalar confidence in $[0, 1]$, returned as a strict JSON object with key `confidence`, conditioned on the source sentence and the model translation rather than the human reference.

All runs use temperature 0 and provider-side output limits fixed in advance within the shared evaluation protocol. These limits serve to standardize response formatting and generation budget across providers rather than to define a substantive experimental variable. In the released artifact, all providers use the same shared prompt renderer for each selected variant (canonical in the bundled runs), with provider-specific adapters handling API transport only. When outputs are malformed or missing required fields, the pipeline applies deterministic extraction and repair heuristics documented in Appendix A and in the accompanying reproducibility package. Because the parse-warning/repair burden differs across providers in the released snapshot, downstream confidence comparisons should be interpreted with that potential confound in mind. Latency and token usage are logged whenever the provider SDK exposes them. Because latency is measured through live provider APIs, it should be interpreted as deployment-facing wall-clock behavior under the recorded runs rather than as a pure model-intrinsic efficiency measure.

3.4 Pipeline and artifacts

The pipeline consists of dataset creation, model calls, feature extraction and metric computation, aggregation, and analysis/plot generation. The accompanying reproducibility package includes prompts, configuration files, generated figures and tables, and the metadata needed to regenerate the reported outputs.

4 Methods

4.1 Reference-based quality

Each translation is scored against the reference using sentence-level chrF as the main quality signal, computed with `sacrebleu`. In the released code path, this is `sacrebleu.sentence_chrf(hypothesis, references=[ref], char_order=6, word_order=2)` (i.e., sentence chrF++ with word n-grams

enabled). chrF is inexpensive, easy to reproduce, and still widely used in WMT-style evaluation. Learned metrics such as COMET, BLEURT, and xCOMET often correlate better with human judgments, especially when finer-grained error analysis is needed. This paper nevertheless uses chrF as the primary metric because the aim is to study cross-provider confidence behavior under a lightweight and fully reproducible reference-based protocol. Sentence-level BLEU is also computed by the pipeline and remains available as a conservative overlap-based secondary check. COMET/xCOMET are not used in the released offline artifact because the required packages and checkpoints are not bundled and would require additional external downloads.

4.2 Source-side surface-complexity proxy

A lightweight source-side surface-complexity score is computed from surface features that do not require model internals:

- **Length:** number of whitespace tokens (`src_len_tokens`);
- **Depth-like heuristic:** a capped function of length ($\texttt{syntactic_depth} \approx \min(10, \lfloor \texttt{len}/4 \rfloor + 1)$);
- **Lexical rarity ratio:** fraction of tokens with length ≥ 8 (`rare_ratio`);
- **Name proxy:** count of capitalized tokens (`ner_count`).

Each feature is z-scored across all sentences and summed:

$$\text{complexity}(x) = z(\text{len}) + z(\text{depth}) + z(\text{rare}) + z(\text{names}).$$

Each example is then assigned to a surface-complexity quartile (Q1 lowest complexity to Q4 highest complexity) using the empirical quartiles of this score. This proxy is intentionally shallow and is used only for stratified descriptive analysis; it is not a validated measure of translation difficulty. In particular, the depth-like term is a coarse length-derived heuristic rather than an independent syntactic analysis.

4.3 Operational low-quality definitions

To study calibration and thresholded mismatch using a reproducible binary target, a low-quality event is defined by a quantile threshold on sentence-level chrF:

- **Within-model low-quality cases:** for each model, mark the lowest 20% chrF examples as low quality.
- **Global low-quality cases:** mark the lowest 20% chrF examples across all models as low quality.

Unless stated otherwise, results use the within-model low-quality definition. Under the current implementation (strictly below a per-model quantile threshold), low-quality prevalence is approximately 20% within each model in this snapshot ($99/500 = 19.8\%$ for each model). The resulting target is therefore a relative quality event rather than an absolute error rate, and it should not be read as a direct measure of semantic incorrectness. The main reason for using this target is comparability across providers under a reproducible reference-based protocol, not the claim that the bottom-20% boundary has intrinsic semantic status.

4.4 Calibration and reliability diagrams

Expected calibration error (ECE) is computed with $B = 10$ confidence bins, where bin accuracy is defined as the fraction of non-low-quality outputs under the chrF-based operational label:

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|S_b|}{N} \left|\mathrm{acc}(S_b) - \mathrm{conf}(S_b)\right|,$$

where $\mathrm{acc}(S_b)$ is the fraction of non-low-quality outputs in bin b and $\mathrm{conf}(S_b)$ is the mean confidence in that bin. In this paper, ECE therefore measures alignment between self-reported confidence and a reproducible chrF-based low-quality event, not direct calibration to semantic correctness. Because the within-model label yields approximately 80% non-low-quality prevalence in this snapshot, ECE should be read as a comparative alignment diagnostic rather than as a complete measure of decision usefulness: a model can look moderately well aligned while still offering limited separation for routing. Recent LLM-specific surveys emphasize that calibration for generated text is harder than for classification because correctness is task-dependent and often only weakly approximated by a single automatic metric. Reliability diagrams are used here primarily as comparative diagnostics across providers under a shared protocol. Because some models place substantial mass near very high confidence, the resulting plots are best read as qualitative comparative summaries rather than as fine-grained probability calibration analyses at every score level.

4.5 High-confidence low-quality mismatch

Mismatch at threshold τ is defined as:

$$\Pr(\text{low-quality} \wedge \text{confidence} > \tau).$$

Strict thresholds such as $\tau = 0.9$ are emphasized because these are the settings in which confidence is often used to bypass human review. Verbalized confidence is especially attractive in black-box settings because it requires no access to logits or hidden states, but recent work also shows that such scores are sensitive to prompt format and calibration strategy. In this paper, mismatch is always relative to the operational low-quality label defined above. This statistic is intentionally policy-oriented rather than a complete selective-prediction evaluation: a low value can arise either because the model separates cases well or because it rarely emits scores above τ. For that reason, mismatch@0.9 is interpreted jointly with the confidence-distribution summaries in Table 1, not as a standalone measure of selective utility. Mismatch is also reported across surface-complexity quartiles.

4.6 Uncertainty estimates

The released reproducibility package also includes 95% bootstrap confidence intervals obtained by resampling sentences with replacement within each model and recomputing mean chrF, ECE, and mismatch at the chosen thresholds. For readability, the main paper tables report point estimates, while the bootstrap intervals remain available in the accompanying materials. The headline comparisons in the main text should therefore be read as descriptive point-estimate comparisons, with the released materials providing additional uncertainty information.

4.7 Supplementary mitigation and metric-robustness analyses

Two supplementary analyses are run from the same repository without additional provider API calls. First, a lightweight post-hoc mitigation stage fits a per-model isotonic calibration map on a

Model	Mean chrF	ECE (within-q20)	Mismatch@0.9	Median conf.
anthropic/claude-haiku-4-5-20251001	57.01	0.094	13.4%	0.92
anthropic/claude-opus-4-6	58.45	0.087	7.6%	0.93
anthropic/claude-sonnet-4-5-20250929	56.89	0.095	12.8%	0.95
gemini/gemini-2.5-flash	54.01	0.148	4.2%	0.73
gemini/gemini-2.5-pro	60.07	0.180	18.2%	1.00
openai/gpt-5-mini	57.09	0.152	12.8%	0.93
openai/gpt-5-nano	55.31	0.076	0.2%	0.73
openai/gpt-5.2	58.87	0.134	15.2%	0.93

Table 1: Main summary table from regenerated code-side outputs (within-model chrF-q20 target).

deterministic hash split of the existing outputs (40% calibration, 60% held-out evaluation) using the same within-model chrF-q20 operational target as the main paper. This stage is intended only as a conservative proof-of-possibility for provider-specific mitigation, not as a claim that a single portable calibration rule exists.

Second, a metric-robustness stage recomputes provider-level ECE and mismatch under an alternative low-quality event derived from the strongest secondary metric available in the current environment. The pipeline attempts COMET first; when COMET is unavailable, it falls back to sentence-level BLEU and records that fallback explicitly in the released artifacts.

Third, the artifact now includes a deterministic semantic-audit scaffold for high-confidence chrF mismatches. The scaffold exports a stratified, multi-provider subset for annotation, provides a fixed label guide (*semantic error*, *acceptable paraphrase*, *metric artifact/unclear*), and supports post-hoc aggregation when annotation CSV files are supplied.

5 Results

Table 1 summarizes point estimates for mean chrF, calibration (ECE), and the rate of high-confidence low-quality outputs at $\tau = 0.9$ under the within-model bottom-20% chrF definition. Mean chrF is fairly clustered (54.0–60.1), but confidence behavior relative to this operational target is not. Gemini 2.5 Pro achieves the highest mean chrF, yet it also has the highest mismatch@0.9 rate (18.2%), driven by near-certain confidence values (median 1.00) even on outputs that fall into the model's low-chrF tail. GPT-5-nano is much more conservative (median 0.73), and that conservatism shows up directly in mismatch@0.9 (0.2%), despite lower average chrF. This does not mean that GPT-5-nano provides the best overall selective-routing signal; rather, it shows that a strict threshold such as 0.9 produces a very different operating point on a conservative model than on an aggressive one. Because the low-quality label is defined within each model, these results should be read as comparisons of how confidence behaves relative to a shared operational target, not as direct measurements of semantic correctness calibration. The most defensible interpretation is therefore comparative and practical: shared confidence thresholds induce materially different operating points across providers under this protocol. The released bootstrap intervals support the same qualitative picture, but they are not displayed inline in the main table.

The example appendix and exported mismatch examples are included to visualize this operational region, not to claim that every listed case is semantically wrong. Some confident "errors" under the chrF-based label are plausible paraphrases that receive low overlap scores.

Model	r(difficulty, conf)	r(difficulty, chrF)	r(conf, chrF)
anthropic/claude-haiku-4-5-20251001	-0.004	0.006	0.257
anthropic/claude-opus-4-6	-0.136	-0.016	0.481
anthropic/claude-sonnet-4-5-20250929	-0.073	0.019	0.455
gemini/gemini-2.5-flash	-0.083	-0.156	0.224
gemini/gemini-2.5-pro	0.004	0.032	0.233
openai/gpt-5-mini	-0.031	0.027	0.041
openai/gpt-5-nano	0.079	0.036	0.207
openai/gpt-5.2	0.103	0.035	0.107

Table 2: Pearson correlations from regenerated analysis outputs.

5.1 Parse-warning and repair burden in the released snapshot

The parse-audit export shows that 443/4000 rows (11.1%) carry explicit parse warnings after deterministic extraction/repair. The burden is highly uneven across models: Gemini 2.5 Flash has 287/500 warning rows (57.4%), whereas all three OpenAI models show 0/500 warning rows in the same snapshot-level warning field. The stricter raw-preview audit also flags malformed preview fragments beyond explicit warning tokens (452/4000 rows, 11.3%), including 9 rows (0.23%) that were malformed in raw previews but not explicitly tagged by the warning field. These quantities do not invalidate the core portability finding, but they are an important implementation-level confound and motivate cautious interpretation of cross-provider confidence comparisons.

5.2 Confidence vs. source-side complexity and quality

Table 2 reports correlations between the source-side surface-complexity proxy, confidence, and chrF. Confidence is only weakly related to the surface-complexity proxy across models, which limits the interpretive weight that can be placed on the proxy. By contrast, some models—notably Claude Opus 4.6 and Claude Sonnet 4.5—show a more noticeable positive relationship between confidence and quality. This trend is directionally consistent with comparatively better alignment for Claude Opus 4.6 in Table 1, while other models with positive confidence–quality correlation still vary in mismatch rate. Because the proxy is shallow and the main evaluation target is operational, these correlations are better read as descriptive context than as evidence for a strong causal account of when confidence should rise or fall. This keeps the provider-level comparison central and the complexity analysis secondary.

5.3 Reliability and mismatch by source-side complexity

The source-side complexity scatter (Figure 1) is included as descriptive context and should be read together with the small correlation magnitudes in Table 2. Reliability diagrams (Figure 2) show that several models assign very high confidence even where empirical non-low-quality rates are lower under the operational chrF-based label. Because the overlaid figure is visually dense, it is best read as qualitative support for the tabulated comparisons rather than as the primary evidence for model ranking. When mismatch@0.9 is broken down by surface-complexity quartile (Figure 3), the regenerated grouped bars in this snapshot are highest in Q1 and then lower (or flat) in later quartiles, with model-dependent magnitude. This pattern is therefore best interpreted as a secondary descriptive tendency under the current proxy rather than as evidence that the proxy robustly captures translation difficulty.

Model	ECE before	ECE after	Mismatch@0.9 before	Mismatch@0.9 after
anthropic/claude-haiku-4-5-20251001	0.110	0.027	14.3%	0.3%
anthropic/claude-opus-4-6	0.084	0.114	9.0%	2.1%
anthropic/claude-sonnet-4-5-20250929	0.087	0.073	12.1%	1.0%
gemini/gemini-2.5-flash	0.152	0.055	4.3%	0.3%
gemini/gemini-2.5-pro	0.175	0.087	17.5%	0.0%
openai/gpt-5-mini	0.128	0.116	10.8%	0.0%
openai/gpt-5-nano	0.091	0.045	0.3%	0.3%
openai/gpt-5.2	0.155	0.047	16.3%	0.0%

Table 3: Held-out isotonic calibration summary from regenerated calibration artifacts.

5.4 Secondary quality–latency view

Figure 4 provides a secondary deployment-oriented view of mean chrF against average latency. It is included as supplementary context for practical trade-offs rather than as a core result about confidence reliability. The main conclusion of the paper comes from calibration and mismatch rather than from the frontier itself.

5.5 Lightweight post-hoc calibration as a provider-specific mitigation

Table 3 reports a supplementary isotonic regression analysis run on a deterministic per-model 40/60 calibration/evaluation split using the already collected confidence outputs and the same within-model chrF-q20 operational target. On the held-out evaluation split, ECE decreases for seven of the eight models; Claude Opus 4.6 is the only case where the held-out ECE estimate worsens slightly. Mismatch@0.9 decreases or stays unchanged for all models, and it falls to zero for Gemini 2.5 Pro, GPT-5-mini, and GPT-5.2 on the held-out portion.

This result is useful but should be interpreted narrowly. It shows that the emitted confidence values are not irredeemably uninformative: lightweight provider-specific remapping can improve alignment to the operational target. At the same time, the fitted mappings differ by model, so the mitigation does not rescue portability. If anything, it reinforces the paper's main claim that confidence should be treated as a provider-specific decision signal rather than as a shared probability scale.

5.6 Metric robustness beyond the main chrF target

Table 4 compares the main chrF-based target with the strongest secondary metric available in the current environment. In this regenerated run, COMET was not installed, so the pipeline fell back to sentence-level BLEU. That fallback is not a stronger learned metric and therefore does not eliminate the paper's metric limitation, but it still provides a conservative check on whether the qualitative provider ordering is specific to chrF alone.

The qualitative picture remains similar under the BLEU-derived target: Gemini 2.5 Pro and GPT-5.2 remain among the most mismatch-prone models at $\tau = 0.9$, Claude Opus 4.6 remains comparatively better aligned than the more aggressive high-confidence systems, and GPT-5-nano remains the most conservative model. The absolute values do move, sometimes non-trivially, which is exactly why the paper continues to frame its claims as protocol- and target-dependent. This BLEU-fallback check only reduces (not resolves) concern that the headline pattern is specific to chrF, and it does not remove the need for stronger learned metrics or human evaluation in future work.

Model	Secondary metric	ECE (chrF-q20)	ECE (secondary-q20)	Mismatc
anthropic/claude-haiku-4-5-20251001	Sentence BLEU fallback	0.094	0.099	
anthropic/claude-opus-4-6	Sentence BLEU fallback	0.087	0.092	
anthropic/claude-sonnet-4-5-20250929	Sentence BLEU fallback	0.095	0.098	
gemini/gemini-2.5-flash	Sentence BLEU fallback	0.148	0.152	
gemini/gemini-2.5-pro	Sentence BLEU fallback	0.180	0.186	
openai/gpt-5-mini	Sentence BLEU fallback	0.152	0.152	
openai/gpt-5-nano	Sentence BLEU fallback	0.076	0.088	
openai/gpt-5.2	Sentence BLEU fallback	0.134	0.145	

Table 4: Robustness comparison between the main chrF-based target and the strongest available secondary metric target (current run: sentence BLEU fallback).

Audit item	Count
Candidate pool (conf>=0.9 and chrF-q20 mismatch)	443
Audit sample rows	48
Valid human labels available	0
Sample provider mix (anthropic/openai/gemini)	12/15/21
Sample bucket mix (Q1/Q2/Q3/Q4)	24/12/11/1
Annotated: semantic error	0
Annotated: acceptable paraphrase	0
Annotated: metric artifact/unclear	0

Table 5: Deterministic semantic-audit scaffold for high-confidence chrF mismatches. Label rows remain zero until annotation CSV files are added under `runs/annotations/semantic_audit/`.

5.7 Semantic-audit scaffold for high-confidence mismatch cases

Table 5 summarizes the new semantic-audit package. The selection is deterministic and designed to support reproducible manual validation of the highest-confidence operational mismatches across providers and complexity quartiles. This does not introduce new semantic findings by itself: in the released snapshot, the table reports candidate/sample coverage and no completed annotation labels. The contribution is an auditable bridge from overlap-defined mismatch events to a concrete human-evaluation workflow.

5.8 External comparator baseline (offline)

Table 6 compares self-reported confidence against a simple model-independent surface proxy that uses only source/hypothesis text and parse-warning metadata. The proxy combines length-ratio consistency, digit consistency, punctuation consistency, and a small parse-warning penalty. It is intentionally lightweight and not intended as a state-of-the-art QE model; the goal is to provide an external reference-free comparator that is fully reproducible from the bundled artifact.

In this snapshot, the comparison is mixed. Self-reported confidence yields lower accepted-error among top-scored outputs for six of eight models, while the surface proxy is stronger for both Gemini models and for some correlation rows. This supports treating self-reported confidence as a useful but not uniformly dominant provider-local signal, while leaving the main non-portability conclusion unchanged.

Model	Corr(self,chrF)	Corr(proxy,chrF)	Accepted err@top20% self	Accepted er
anthropic/claude-haiku-4-5-20251001	0.257	0.239	0.120	(
anthropic/claude-opus-4-6	0.481	0.404	0.060	(
anthropic/claude-sonnet-4-5-20250929	0.455	0.465	0.110	(
gemini/gemini-2.5-flash	0.224	0.684	0.130	(
gemini/gemini-2.5-pro	0.233	0.124	0.220	(
openai/gpt-5-mini	0.041	0.102	0.130	(
openai/gpt-5-nano	0.207	0.060	0.110	(
openai/gpt-5.2	0.107	0.053	0.160	(

Table 6: External comparator against self-reported confidence using a model-independent surface proxy (higher correlation and lower accepted-error are better).

Prompt variant	Role	Status	Models with data
canonical_v1	baseline	available_baseline	8
minimal_v2	variant	missing_optional_live_run	0
verifier_v3	variant	missing_optional_live_run	0

Table 7: Prompt-sensitivity availability in the released artifact bundle. Non-baseline variants require optional live reruns and are not fabricated offline.

5.9 Prompt-sensitivity robustness status

The codebase now defines three schema-equivalent prompt variants (`canonical_v1`, `minimal_v2`, `verifier_v3`) that keep task and JSON schema fixed while varying phrasing. The released offline snapshot includes only baseline (`canonical_v1`) outputs. Non-baseline variants are optional live reruns (credentials required) and are not fabricated in this artifact.

Table 7 reports availability status. Because non-baseline variant outputs are absent in the bundled snapshot, this paper does not claim cross-prompt invariance.

6 Discussion and limitations

Self-reported confidence scales are not directly comparable across providers. Confidence distributions differ sharply (Table 1). Gemini 2.5 Pro often outputs values at or near 1.00, whereas GPT-5-nano and Gemini 2.5 Flash are more conservative (median 0.73). As a result, a fixed threshold policy such as "route to a human if confidence < 0.9" would behave very differently across systems. Without calibration or per-provider tuning, that policy is not portable. This is the paper's main positive result. At the same time, low mismatch at a strict threshold should not be over-read as proof of a globally better confidence signal, because conservatism can also reduce the joint event $\Pr(\text{low-quality} \wedge \text{confidence} > \tau)$. Recent work on verbalized confidence supports this interpretation: black-box confidence scores can be informative, but they are highly sensitive to elicitation and often need explicit post-hoc calibration [2].

The central construct is operational rather than semantic. The paper evaluates self-reported confidence against a chrF-based low-quality label rather than against human judgments of translation correctness. Under the within-model definition in the released implementation, low-quality prevalence is approximately 20% (19.8% per model in this snapshot), so the analysis compares how confidence aligns with a relative quality event within each model. This makes the

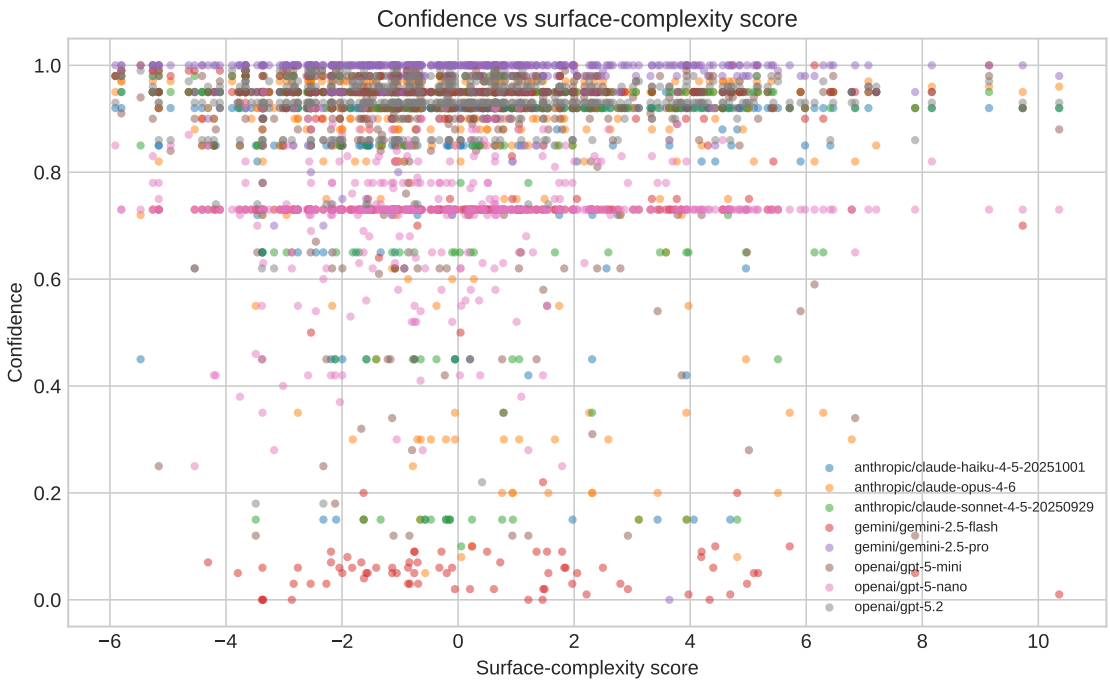

Figure 1: Source-side surface-complexity proxy vs. self-reported confidence. The plot provides descriptive context only; the associated correlations in Table 2 are more interpretable than the raw scatter.

results useful for controlled cross-provider comparison, but it also limits how strongly they can be interpreted as evidence about semantic correctness calibration. The manuscript is therefore strongest when it restricts its claim to portability under this protocol and target rather than to confidence faithfulness in a broader sense.

The source-side surface-complexity analysis should be read cautiously. Correlations between the source-side surface-complexity proxy and confidence are small (Table 2), which is consistent with the idea that reported confidence is not derived directly from these surface features. In this regenerated snapshot, the quartile-based mismatch plots show a consistent Q1-heavy descriptive pattern, but this remains a proxy-dependent stratification result rather than a validated difficulty effect. The proxy is therefore useful mainly for coarse stratification, not for strong claims about translation difficulty. This is also a useful multilingual caution: recent confidence-estimation work shows that reliability can vary substantially across languages and prompting conditions, even when the same model family is used.

Robustness to the low-quality definition. The main analysis defines low-quality cases as within-model bottom 20% chrF. Appendix C shows that mismatch@0.9 remains qualitatively similar under the chrF-based low-quality definitions available in the released snapshot used for this paper (within-model bottom 20% chrF and global bottom 20% chrF). The ranking is not identical under every choice, but the qualitative gap between conservative and more aggressive confidence behavior remains.

Prompt robustness remains an explicit open check in this snapshot. Prompt-variant scaffolding is now integrated into the execution code and configuration, but only baseline-variant outputs are bundled offline. The central portability claim is therefore evidenced for the released

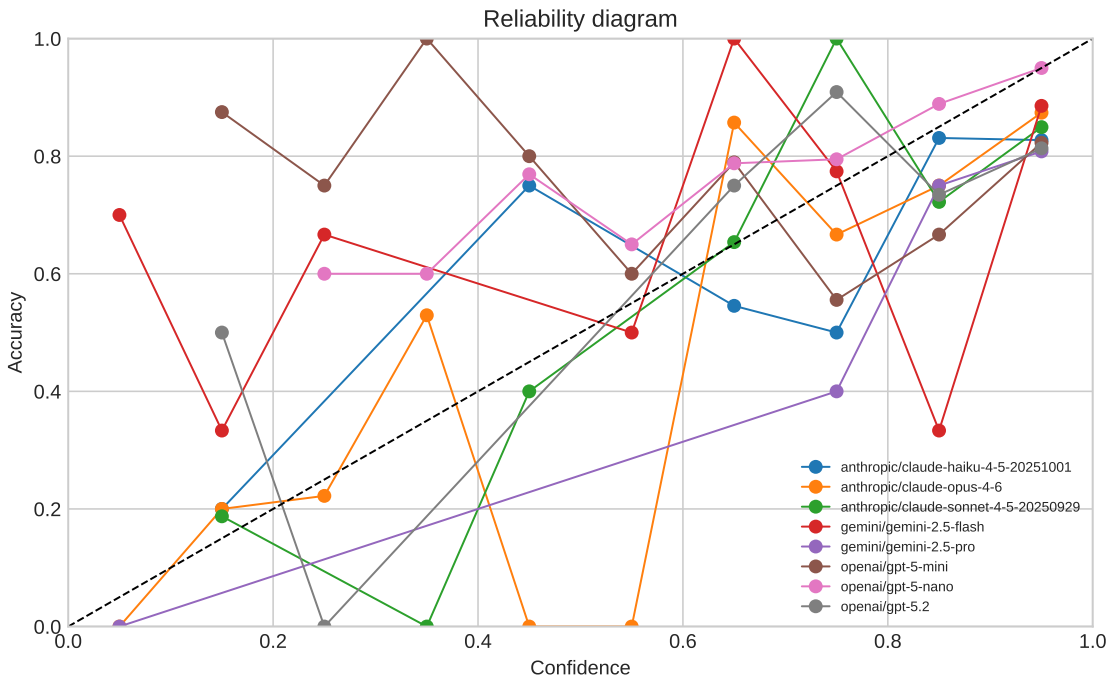

Figure 2: Overlaid reliability diagram with 10 bins. This figure is intended as a qualitative comparative view relative to the operational low-quality label rather than as the primary evidence for model ranking.

baseline protocol; cross-prompt stability remains to be tested via optional live reruns.

What the supplementary mitigation does and does not show. The isotonic post-hoc analysis is encouraging in one limited sense: on a held-out split, provider-specific remapping often reduces ECE and lowers mismatch@0.9 relative to the same operational target. That makes provider-side mitigation more plausible for deployment. It does not, however, make the confidence scale portable. The learned remapping is model-specific, the gains are uneven, and one model (Claude Opus 4.6) shows worse held-out ECE after calibration. The right takeaway is therefore not that a lightweight calibration stage solves the problem, but that the problem can sometimes be mitigated only after treating each provider separately.

Practical implication. The strongest practical implication is not that self-reported confidence is useless, but that it should be treated as a provider-specific decision signal rather than as a portable probability. In settings where confidence is used for routing or abstention, the immediate remedies suggested by the present results are provider-specific threshold tuning, explicit post-hoc calibration, or the addition of an external evaluator layer rather than blind reuse of a shared confidence threshold across APIs. A stronger selective-prediction evaluation would require additional utility-oriented metrics or threshold sweeps, but the present evidence is already sufficient to reject the idea that one strict threshold transfers cleanly across providers in this setup.

6.1 Limitations and threats to validity

Self-reported confidence. The confidence signal comes from black-box APIs and may reflect provider-specific conventions or instruction-following behavior rather than calibrated probabilities.

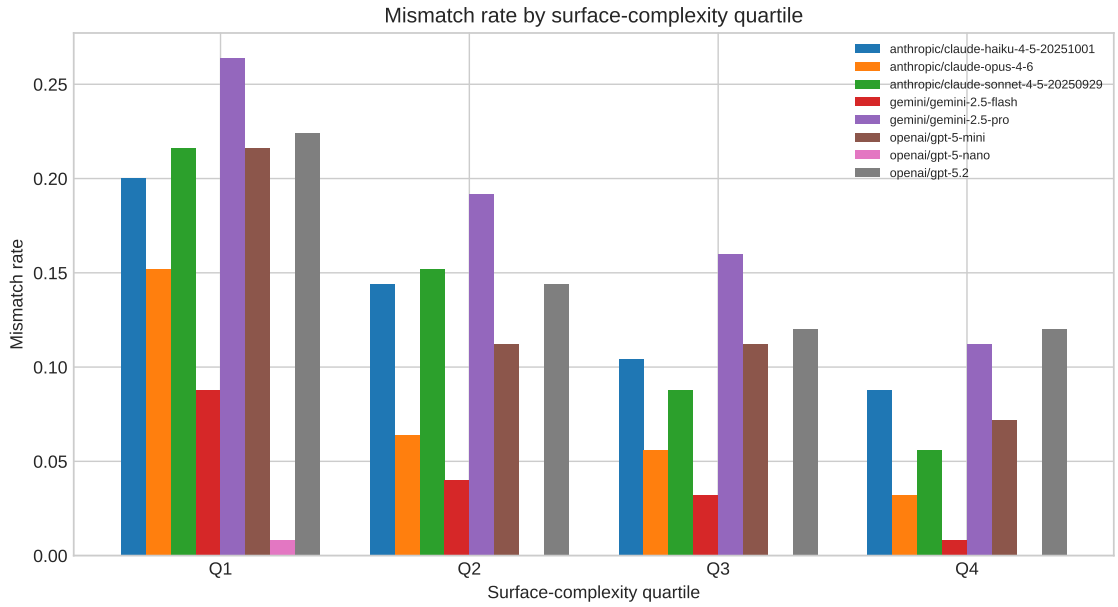

Figure 3: Mismatch@0.9 by source-side surface-complexity quartile. The figure is best interpreted as a secondary stratified view rather than as evidence for a strong difficulty effect.

Because confidence is elicited after generating the translation and conditioned on that translation, it may also include post-hoc rationalization.

Automatic metrics as a proxy for correctness. The paper evaluates confidence against chrF-based operational low-quality labels rather than human judgments of correctness. These labels can disagree with human judgments and can overweight surface similarity. Stronger learned metrics such as COMET, BLEURT, and xCOMET often provide a better approximation to human assessment. The supplementary BLEU-based robustness stage reduces this limitation only partially, and the new semantic-audit scaffold is currently infrastructure without completed annotations in the released bundle. Human labeling or a bundled learned-metric checkpoint would still materially strengthen the claims.

Selective utility is only partially observed. ECE and mismatch@0.9 help characterize operating points under the chosen target, but they do not by themselves provide a full selective-prediction evaluation. In particular, low mismatch at a strict threshold can reflect either useful separation or simple reluctance to emit very high confidence. A fuller routing analysis would require additional coverage-aware metrics or threshold sweeps.

Protocol and scope dependence. The study covers one language pair (English-to-German), one dataset (WMT17), one prompt protocol, one decoding/setup configuration, and 500 sampled sentences from a released snapshot. Generalization to other language pairs, domains, prompt variants, model revisions, and longer segments remains open.

The source-side surface-complexity proxy is intentionally simple. The proxy uses basic source-side features rather than a semantic or human-judged notion of difficulty. More linguistically informed features, learned QE signals, or human annotations could provide sharper stratification.

Prompt sensitivity and parse/repair burden. Prompts are standardized and temperature 0 is used throughout, but confidence elicitation can still be sensitive to wording and formatting constraints. In this released snapshot, 443/4000 rows (11.1%) carry parse warnings, with substantial model-level spread (e.g., 287/500 for Gemini 2.5 Flash vs. 0/500 for each OpenAI model in the warning field). The stricter raw-preview audit flags 452/4000 rows (11.3%),

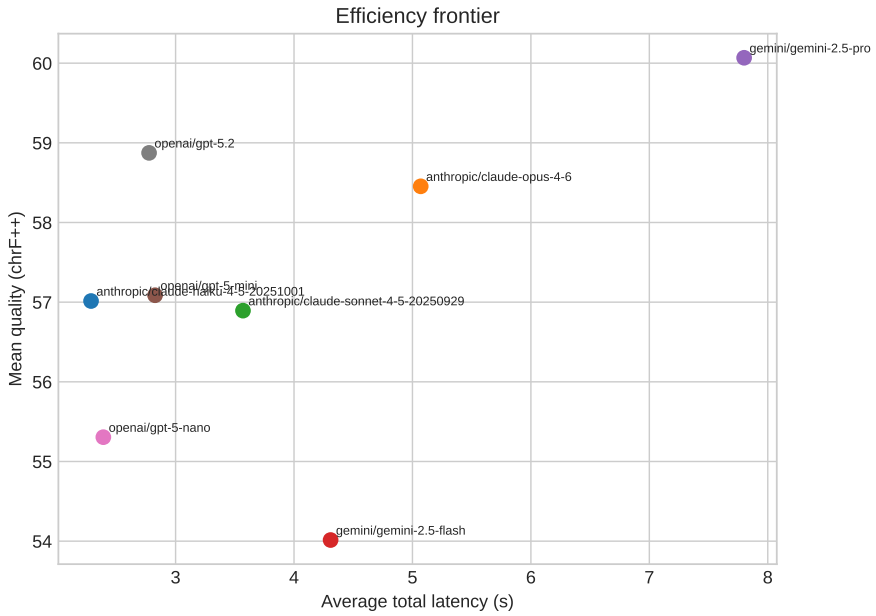

Figure 4: Secondary quality–latency view across models. This figure provides supplementary deployment context and is not central to the paper's main claim about confidence portability.

including 9 unflagged malformed preview cases. Deterministic repair/fallback logic is applied before aggregation, and this may itself affect measured confidence distributions, calibration, and mismatch rates. This is consistent with recent evidence on verbalized confidence in LLMs [3]. Systematic prompt-robustness experiments remain future work.

7 Conclusion

A standardized confidence-elicitation protocol was run for English-to-German MT across eight LLMs from three providers. Although mean chrF scores are relatively similar across models, ECE and the prevalence of high-confidence low-quality outputs vary widely relative to the operational chrF-based target used here. The strongest result is therefore comparative and practical: under this prompt protocol and evaluation setup, self-reported confidence should not be treated as a portable probability across providers, and a shared threshold does not produce comparable operating points across systems. This conclusion is about portability under an operational overlap-based label; it does not establish semantic calibration.

The source-side surface-complexity analysis provides only a coarse secondary view. Its relationship to confidence is weak, and it should not be interpreted as a validated measure of translation difficulty. The supplementary isotonic analysis shows that provider-specific post-hoc remapping can improve alignment to the operational target for most models, but that mitigation is itself non-portable and therefore does not change the central conclusion. The secondary BLEU-based robustness stage yields a broadly similar cross-provider picture, while the added semantic-audit scaffold makes targeted human follow-up reproducible but does not itself add semantic labels in this release. A simple external comparator baseline suggests that self-reported confidence has useful signal relative to a lightweight model-independent proxy, but this does not rescue portability across providers. Prompt-variant infrastructure is now included, yet non-baseline variant outputs are not bundled offline, so cross-prompt invariance is not claimed here. The most plausible next step motivated by these results is therefore to treat confidence as a signal that may still be useful, but only after provider-specific threshold tuning, explicit post-hoc calibration,

or combination with stronger external evaluation layers. Recent work on uncertainty-aware MT evaluation and quality estimation points in this direction, including LLM-based MT evaluation, black-box confidence calibration from generations alone [2], and confidence-aware QE models such as Early-Exit and Instant Confidence COMET. The present results do not establish that the most conservative model is the most useful in general; they establish that fixed-threshold operating points are provider-dependent under the released protocol and target.

Reproducibility

A reproducibility package accompanies the paper. Repository: `https://github.com/altieriumb7/llm-confidence-portability-mt.git`. The reported analysis was regenerated from the released repository snapshot with seed **123** on the WMT17 `en-de` test set sampled to **500** sentence pairs via `sacrebleu -echo`. The aggregated dataset contains **4,000** model outputs (**500/500** for each of eight models). The accompanying materials record the repository commit, configuration hash, and analysis timestamp used to regenerate the paper outputs. The same bash entrypoint can also run the supplementary calibration, parse-audit, secondary-metric, and metric-robustness stages without new provider API calls when the required artifacts already exist. End-to-end scripts, prompts, and generated tables and figures are included.

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[3] Gal Yona, Roee Aharoni, and Mor Geva. Can large language models faithfully express their intrinsic uncertainty in words? In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 7752–7764, Miami, Florida, USA, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.443. URL `https://aclanthology.org/2024.emnlp-main.443/`.

A Prompt templates used in the canonical runs

All providers are instructed to return strict JSON only. In the released artifact, prompts for each variant are rendered from shared templates and reused across provider adapters, so the wording below is the canonical wording used for all providers in bundled runs. When provider outputs are malformed or missing required fields, the pipeline applies deterministic extraction and repair heuristics before downstream aggregation.

A.1 Translation prompt

```
SYSTEM: You are a helpful assistant. Output must be strict JSON.
USER: Translate English to German. Return ONLY one JSON object with exactly
this key: {"translation":"..."}. No explanations, no markdown, no code fences.
SOURCE: <English sentence>
```

A.2 Confidence prompt

The prompt asks for "translation correctness confidence" because that is the elicited verbalized signal. The paper then evaluates that signal against an operational chrF-based low-quality label rather than against a human correctness judgment.

```
SYSTEM: You are a helpful assistant. Output must be strict JSON.
USER: Rate translation correctness confidence from 0 to 1. Return ONLY one
JSON object with exactly this key: {"confidence":0.73}. No explanations, no
markdown, no code fences.
SOURCE: <English sentence>
TRANSLATION: <German hypothesis>
```

B Illustrative operational mismatch examples

The examples below are illustrative only. They are not used to establish semantic error and should not be read as evidence of semantic incorrectness by themselves. Overlap-based metrics can under-reward semantically adequate paraphrases, especially at the sentence level. These examples are therefore included only to visualize the high-confidence, low-chrF region under the paper's operational definitions. The full exported list is provided in `paper/top_mismatch_examples.md`.

id	bucket	conf	chrF	example (source / hypothesis / reference)
110	Q1	0.99	16.06	However, volunteers are already aware of this. / Freiwillige sind sich dessen jedoch bereits bewusst. / Sensibilisiert seien die Helfer aber schon.
11	Q1	0.98	33.49	"The benefits of the restructuring programme will only become visible after 5 years," he said. / "Die Vorteile des Umstrukturierungsprogramms werden erst nach fünf Jahren sichtbar werden", sagte er. / "Die Früchte der Umstrukturierung sind aber auch nach 5 Jahren noch nicht zu sehen", hieß es.
266	Q1	0.99	39.95	It is still apparently unclear who is behind the attacks. / Es ist offenbar immer noch unklar, wer hinter den Angriffen steckt. / Noch sei unklar, welche Gruppe hinter den Anschlägen stecke.

Table 8: Illustrative high-confidence, low-chrF cases from the GPT-5.2 export. These examples visualize the operational mismatch region used in the paper, but they are not evidence of semantic incorrectness by themselves and may include semantically acceptable paraphrases that receive low overlap scores.

C Robustness analyses

Table 9 reports mismatch@0.9 under the two chrF-based low-quality definitions available in the released snapshot used for this paper: (i) within-model bottom 20% chrF and (ii) global bottom 20% chrF across all models. The ranking is not identical under the two definitions, but the broad separation between more conservative and more aggressive confidence behavior remains.

Model	Mismatch@0.9 (within-model chrF-q20)	Mismatch@0.9 (global chrF-q20)
anthropic/claude-haiku-4-5-20251001	13.4%	13.4%
anthropic/claude-opus-4-6	7.6%	6.4%
anthropic/claude-sonnet-4-5-20250929	12.8%	14.2%
gemini/gemini-2.5-flash	4.2%	5.6%
gemini/gemini-2.5-pro	18.2%	12.6%
openai/gpt-5-mini	12.8%	14.6%
openai/gpt-5-nano	0.2%	0.8%
openai/gpt-5.2	15.2%	11.8%

Table 9: Appendix robustness comparison across chrF-based low-quality definitions at $\tau = 0.9$.

Table 4 adds a second robustness view based on the strongest secondary metric available in the current environment. In the present regenerated run, COMET was unavailable and the pipeline fell back to sentence-level BLEU, so this appendix should be read as a conservative overlap-based cross-check rather than as learned-metric validation. Table 5 then adds the semantic-audit scaffold summary for follow-up human review. Table 6 provides an external comparator baseline from a model-independent surface proxy, and Table 7 records prompt-variant availability status in the released artifact.